BAM as a Transformer add-on

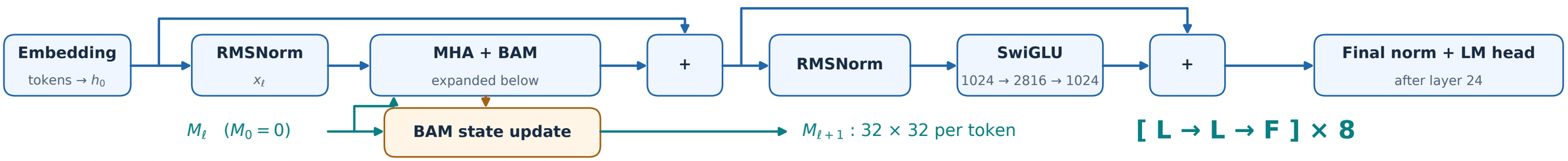
BamLlama2MediumV2C256LocalFetchC8SharedReadLLFScan

24 layers / D = 1024 / 16 MHA heads × 64 / SwiGLU width = 2816

Transformer · BAM read / route · BAM write

01 / MODEL & DECODER

Two streams through depth: residual vectors + per-token matrix state



Repeat the decoder from RMSNorm to the second residual add; each layer has its own parameters.

02 / LOCAL LAYER

L · Shared read into V and O

Layers 1, 2, 4, 5, ..., 22, 23

$x_\ell = \mathrm{RMSNorm}(h_\ell)$ · $M_\ell(t) \in \mathbb{R}^{32\times32}$

Q / K projections + RoPE
$Q = \mathrm{RoPE}(xW_Q) + \Delta Q$
$K = \mathrm{RoPE}(xW_K) + \Delta K$

$\Delta Q, \Delta K$

Local Q/K read
Full M · rank-1 head routing
Keys, gates & head weights from x

Causal attention weights
$A_n = \mathrm{softmax}(Q_n K_n^T/\sqrt{64} + \mathrm{mask})$

Address-axis compression
$S_\ell = M_\ell P_c$ (32×8)

$V^0 = xW_V$

One bilateral read
$R_n(t) = \mathrm{Read}(S_\ell(t), x_t)$
RMS keys; shared pre-gate result

MHA value aggregation
$Y_n(t) = \sum_{s\le t} A_n(t,s)\,[V_n^0(s) + \Delta V_n(s)]$

$G_V(x) \odot R$ — ΔV

$G_O(x) \odot R$ — ΔO

Fused head output
$O_n = Y_n + \Delta O_n$

$\mathrm{Concat}(O_n)\,W_O$
→ residual add → RMSNorm → SwiGLU → add

O → matrix write (below)

Shared read; independent V/O gates.

Each gate has separate row / column factors.

03 / FETCH LAYER

F · MHA routes matrix retrieval

Layers 3, 6, 9, ..., 24

$x_\ell = \mathrm{RMSNorm}(h_\ell)$ · $M_\ell(t) \in \mathbb{R}^{32\times32}$

Q / K projections + RoPE
$Q = \mathrm{RoPE}(xW_Q) + \Delta Q$
$K = \mathrm{RoPE}(xW_K) + \Delta K$

$\Delta Q, \Delta K$

Local Q/K read
Full M · rank-1 head routing
Keys, gates & head weights from x

Causal attention weights
$A_n = \mathrm{softmax}(Q_n K_n^T/\sqrt{64} + \mathrm{mask})$

Address-axis compression
$S_\ell = M_\ell P_c$ (32×8)

$V^0 = xW_V$

Reuse MHA attention weights
$\beta_{ts} = \sum_n w_{tn}(x_t) A_n(t,s)$
Signed head mix; set $\beta_{tt} = 1$

MHA value aggregation
$Y_n(t) = \sum_{s\le t} A_n(t,s) V_n^0(s)$

Fetch compressed matrices
$\bar{S}_t = \sum_{s\le t} \beta_{ts} S_\ell(s)$

Fused head output
$O_n = Y_n + \Delta O_n$

Bilateral read + gate
$\Delta O_n = \mathrm{GatedRead}_n(\bar{S}_t, x_t)$

$\mathrm{Concat}(O_n)\,W_O$
→ residual add → RMSNorm → SwiGLU → add

One fetched matrix; 16 output read heads.

O → matrix write (below)

04 / WRITE IN EVERY LAYER

The fused MHA + BAM head output becomes new matrix content

Data from fused head output
$u_n = \mathrm{RMS}(O_n[0:32])$

Address + write gate from x
$p_n = \mathrm{RMS}(P_{loc,n}(x))$, $g_n = \sigma(W_{g,n}x + b_n)$

Per-token outer-product update
$M_{\ell+1}(t) = M_\ell(t) + \sum_{n=1}^{16} g_{tn} u_{tn} p_{tn}^T$

To the next layer
Full 32 × 32 state

Local = same-token BAM read. All 24 layers use global causal MHA. C256 = query tiling; C8 = compressed address width.

Layer scan executes [L,L,F] blocks with distinct layer parameters. Q/K additions are after RoPE; QK normalization is disabled.